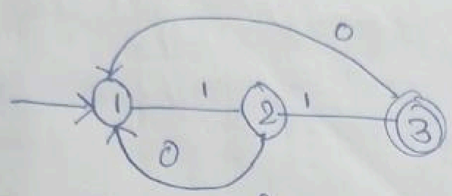
8/07/2024 Assignment - 5

(1)



using Arden's theorem and kleen closure theorem.

using Arden's theorem

$(q_1) \Rightarrow q_2 \cdot 0 + \varepsilon$ — (1)

$(q_2) \Rightarrow q_1 \cdot 1$ — (2)

$(q_3) \Rightarrow q_2 \cdot 1 + q_1 \cdot 0$ — (3)

$$q_1 \Rightarrow \underset{R}{Q_2} \cdot \underset{P}{0} + \underset{Q}{\varepsilon}$$

$$\Rightarrow \varepsilon (0)^*$$

$$q_2 \Rightarrow \varepsilon (0)^* \cdot 1$$

$$q_3 \Rightarrow \varepsilon (0)^* \cdot 1 \cdot 1 + \varepsilon (0)^* \cdot 0$$

using kleen closure theorem

$$R_{ij}^{k} = R_{ij}^{k-1} + R_{ik}^{k-1} (R_{kk}^{k-1})^* (R_{kj}^{k-1})$$

initial $= 1 = i$

final $= 3 = j$

$k = 3$

$$R_{13}^{3} = R_{13}^{2} + R_{13}^{2} (R_{33}^{2})^* (R_{33}^{2})$$

Now, to find $R_{13}^{2} \Rightarrow i = 1 \quad j = 3 \quad k = 2$. to find R_{13}^{2}

$R_{13}^{2} \Rightarrow i=1 \; ; \; j=3 \; ; \; k=2$

$R_{13}^{2} \Rightarrow R_{13}^{1} + R_{12}^{1} \, (R_{22}^{1}) \, R_{23}^{1}$

let $k=1$ to find $R_{12} ; R_{13} ; R_{22} ; R_{33}$

$R_{12}^{1} \Rightarrow R_{12}^{0} + R_{11}^{0} (R_{11}^{0})^{*} R_{12}^{0} \Rightarrow 1 + (\epsilon)^{*} . 1$

$R_{13}^{1} \Rightarrow R_{13}^{0} + R_{11}^{0} (R_{11}^{0})^{*} R_{13}^{0} \Rightarrow 0 . (\epsilon)^{*} . 0$

$R_{22}^{1} \Rightarrow R_{22}^{0} + R_{21}^{0} (R_{11}^{0})^{*} R_{12}^{0} \Rightarrow \epsilon + (\epsilon)^{*} . 1 . 0 \Rightarrow \epsilon + 0.1$

$R_{33}^{1} \Rightarrow R_{33}^{0} + R_{31}^{0} (R_{11}^{0})^{*} R_{13}^{0} \Rightarrow \epsilon + (\epsilon)^{*} . 0 . \phi \Rightarrow \epsilon + \phi$

$R_{23}^{1} \Rightarrow R_{23}^{0} + R_{21}^{0} (R_{11}^{0})^{*} R_{13}^{0} \Rightarrow 1 + 0 (\epsilon)^{*} . 0 \Rightarrow 1 + 0 . 0$

$R_{32} \Rightarrow R_{32}^{0} + R_{31}^{0} (R_{11}^{0})^{*} R_{12}^{0} \Rightarrow \phi + \phi (\epsilon)^{*} 1 \Rightarrow \epsilon + \phi$

let $k=3$ to find R

$\therefore R_{33}^{2} \Rightarrow R_{33}^{1} + R_{32}^{1} (R_{22}^{1})^{*} R_{23}^{1}$

$R_{12}^{1} \Rightarrow (1+1)$

$R_{13}^{1} \Rightarrow (0+0)$

$R_{22}^{1} \Rightarrow (\epsilon + 1.0)$

$R_{23}^{1} \Rightarrow (1+0.0)$

$R_{33}^{1} \Rightarrow (\epsilon + \phi)$

$R_{32}^{1} \Rightarrow (\phi + \phi)$

$R_{23}^{1} \Rightarrow (1+0.0)$

$\therefore R_{33}^{2} \Rightarrow R_{33}^{1} + R_{32}^{1} (R_{22}^{1})^{*} R_{23}^{1} \longrightarrow$ (2)

$\Rightarrow (\epsilon + \phi) + (\phi + \phi)(\epsilon + 1.0)^{*}(1 + 0.0)$

$\therefore R_{13}^{2} \Rightarrow R_{13}^{1} + R_{12}^{1} (R_{22}^{1})^{*} R_{23}^{1} \longrightarrow$ (3)

$\Rightarrow (0+0) + (1+1)(\epsilon + 1.0)^{*}(1+0.0)$

$\therefore R_{13}^{3} \Rightarrow R_{13}^{2} + R_{13}^{2} (R_{33}^{2})^{*} R_{33}^{2}$

$\Rightarrow [0+0+(1+1)(\epsilon+1.0)^{*}(1+0.0) + [[(0+0)+(1+1)(\epsilon+1.0)^{*}(1+0.0)][(\epsilon+\phi)+(\phi)(\epsilon+1.0)^{*}(1+0.0)]^{*}[(\epsilon+\phi)+\phi(\epsilon+1.0)^{*}(1+0.0)]]$